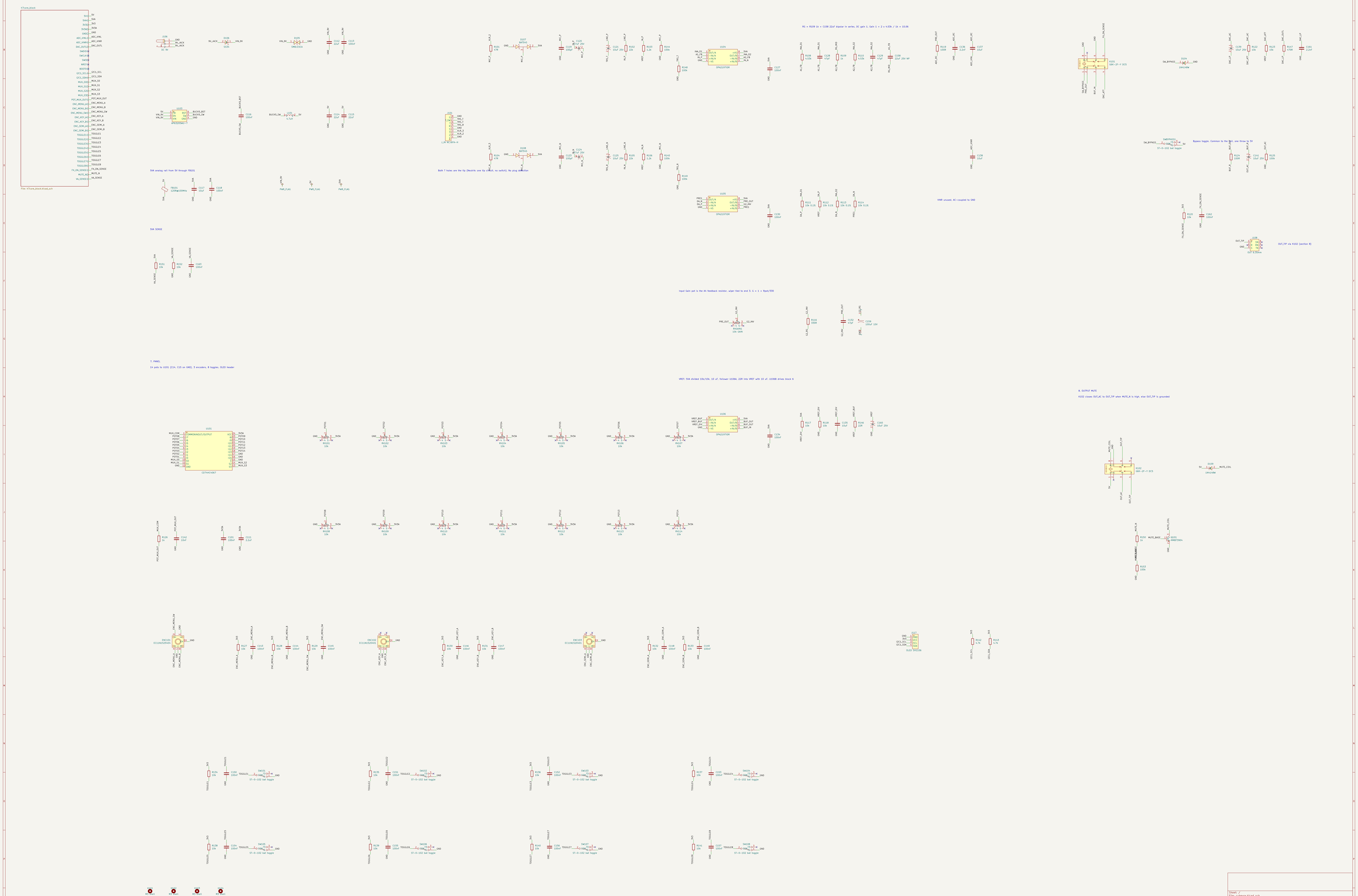
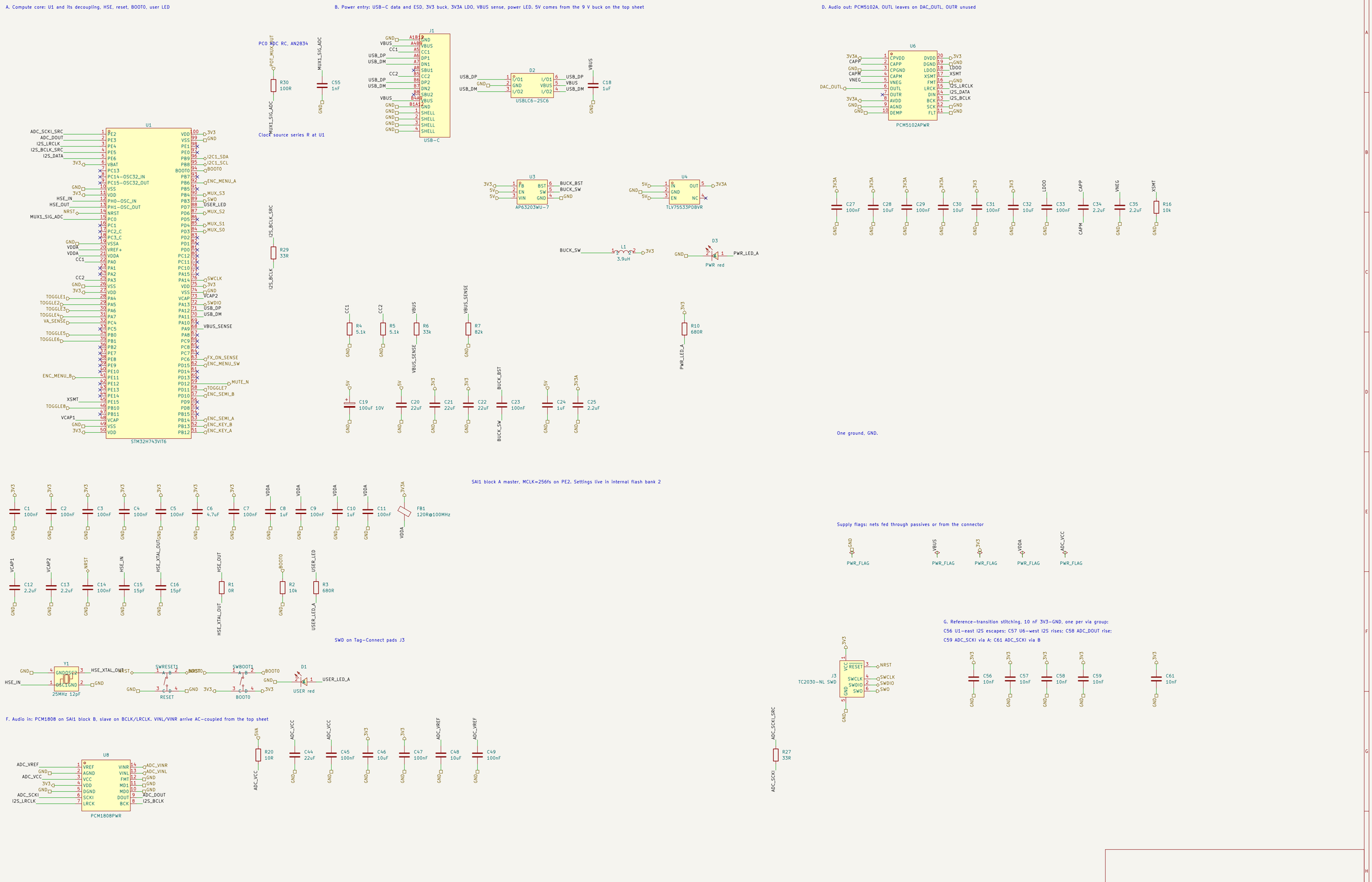

1. POWER IN
9 V Boss supply, centre pin GND. SS3H, TVS, 5 V buck, 5VA rail
2. MIC / LINE INPUT
XLR leg k7R, BAT54S clamp, 100pF, 100k to GND, k7uF + 1k,k, TRS leg 10uF = LINE_n, 22k pad, 100k to GND, 1.2k bias to VREF
3. PREAMP
Stage 1 INA 20.05 dB (U104, U105A). Stage 2 0–29.9 dB on Input Gain pot (U105B)
Chain 20–50 dB XLR / ~5.6 to +24.3 dB TRS; DC gain 1; DC-coupled legs)
RG = R109 1k + C158 22uF bipolar 1n series, DC gain 1. Gain 1 = 2 + k.51k / 1k = 10.06
4. ADC FEED
PRE_OUT always feeds the ADC
5. BYPASS
Pole A COM BUF_IN, NC PRE_OUT, NO DAC_ATT. Pole B COM GND, NO FX_ON_SENSE (low = effect on). Coil on = effect on
6. DAC / OUTPUT
k70R / 2.2uF filter, 10uF, DAC_ATT = 0.6 x DAC about VREF
U108B buffers to the jack
Both T holes are the tip (Neutrik: one tip contact, no switch). No plug detection
Input Gain pot is the A4 feedback resistor, wiper tied to end 3. G = 1 + Rpot/330
VNR unused, AC-coupled to GND
Bypass toggle. Common to the tip, one throw to 5V
OUT_TIP via K102 (section 8)
VREF: 5VA divided 10k/10k. 10 uF follower U106A. 22R into VREF with 10 uF. U106B drives block 6
8. OUTPUT MUTE
K102 closes OUT_AC to OUT_TIP when MUTE_N is high, else OUT_TIP is grounded
7. PANEL
14 pots to U101 (C14, C15 on GND), 3 encoders, 8 toggles, OLED header
5VA analog rail from 5V through FB101
5VA SENSE
k7uF
Sheet: /
File: cubevox.kicad_sch
Title: cubevox vocal FX box
Size: A0
Date: 2026-10-02
Rev: 1
KiCad 10.0.3
5V / 2.7A

A. Compute core: U1 and its decoupling, HSE, reset, BOOT0, user LED



Sheet: /h7core_block/
File: h7core_block.kicad_sch

Title: cubevox H7 core block

Size: A2	Date: 2026-10-02	Rev: 1
KiCad E.D.A. 10.0.5		Id: 2/2